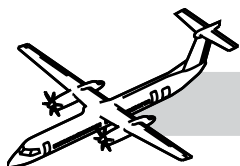


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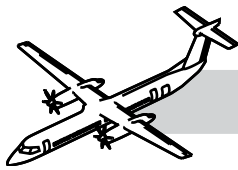
ENGINE INDICATING

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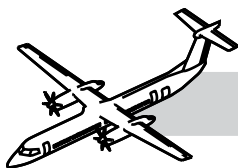


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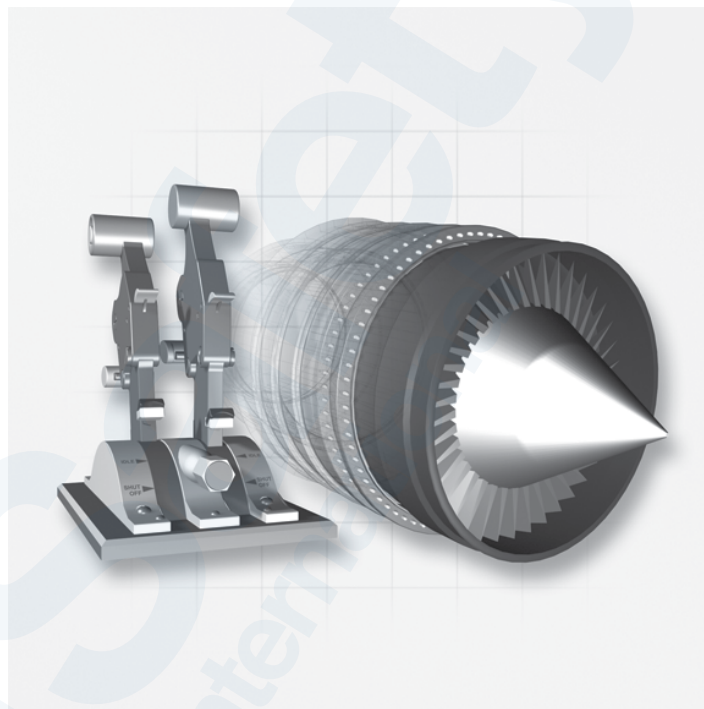
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CHAPTER 77

ENGINE INDICATING



77-00-00 INTRODUCTION

The indicating system gives data on vital engine parameters for use by the flight crew and maintenance personnel to monitor engine health and performance.

The system is comprised of the following sub systems:

- Engine Power indicating
- Engine Temperature Indicating
- Engine Monitoring System.

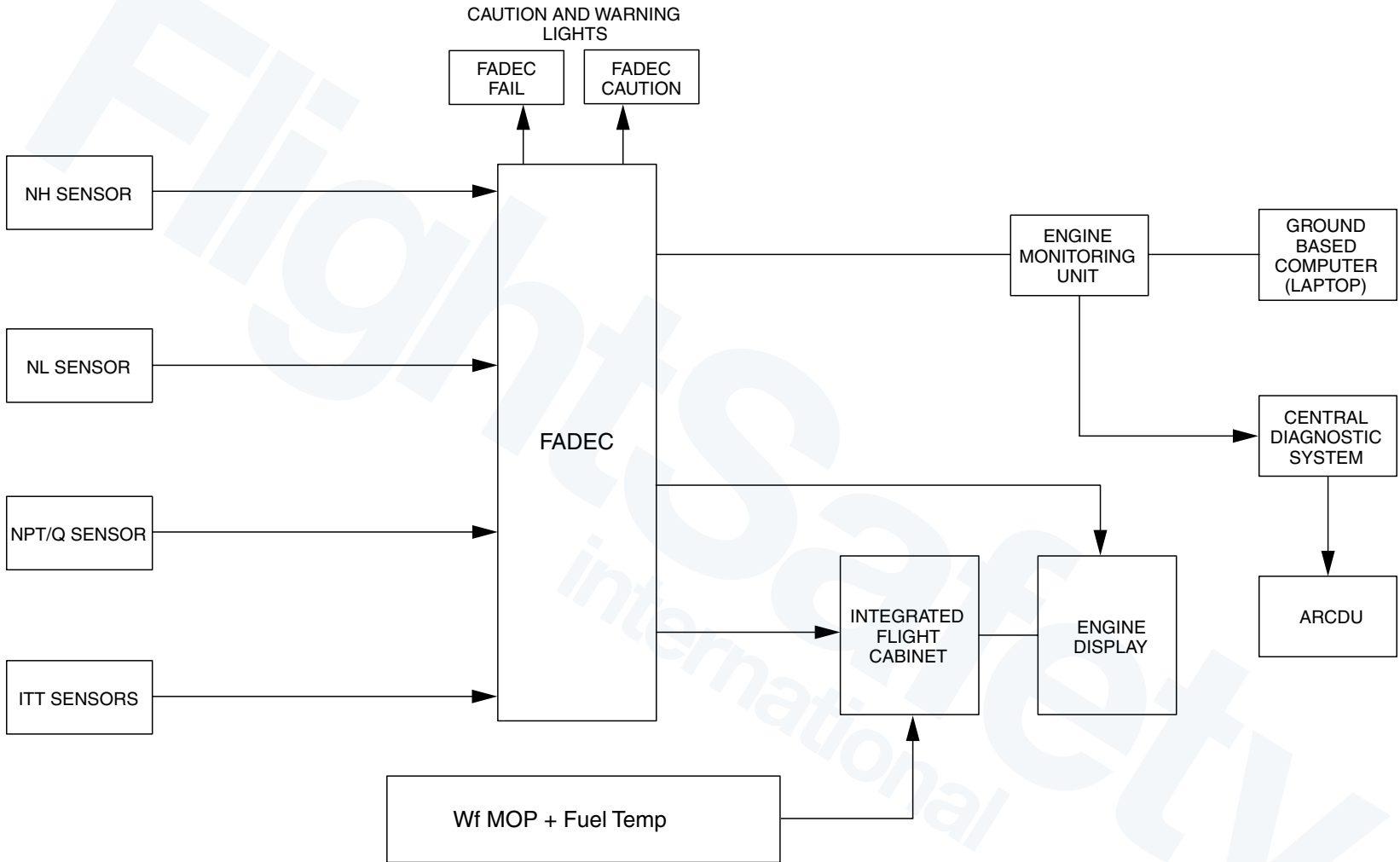
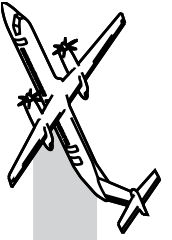
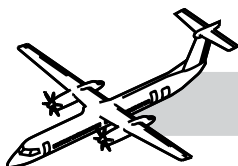


Figure 77-1. Engine Indicating Block Diagram



GENERAL

NOTES

The Monitored engine parameters are:

- High Pressure Rotor Speed (N_h)
- Low Pressure Rotor Speed (N_L)
- Propeller Speed (N_P)
- Torque (Q)
- Air Intake Temperature ($T_{I.8}$)
- Indicated Turbine Temperature (ITT).

Refer to Figure 77-1. Engine Indicating Block Diagram.

The engine parameters from the sub-systems are shown on the ED.

The parameters are also sent to the EMU for use by maintenance personnel.

These parameters are accessed on the maintenance screen (ARCDU) or downloaded and read from the EMU to a computer, equipped with the appropriate Pratt & Whitney software.

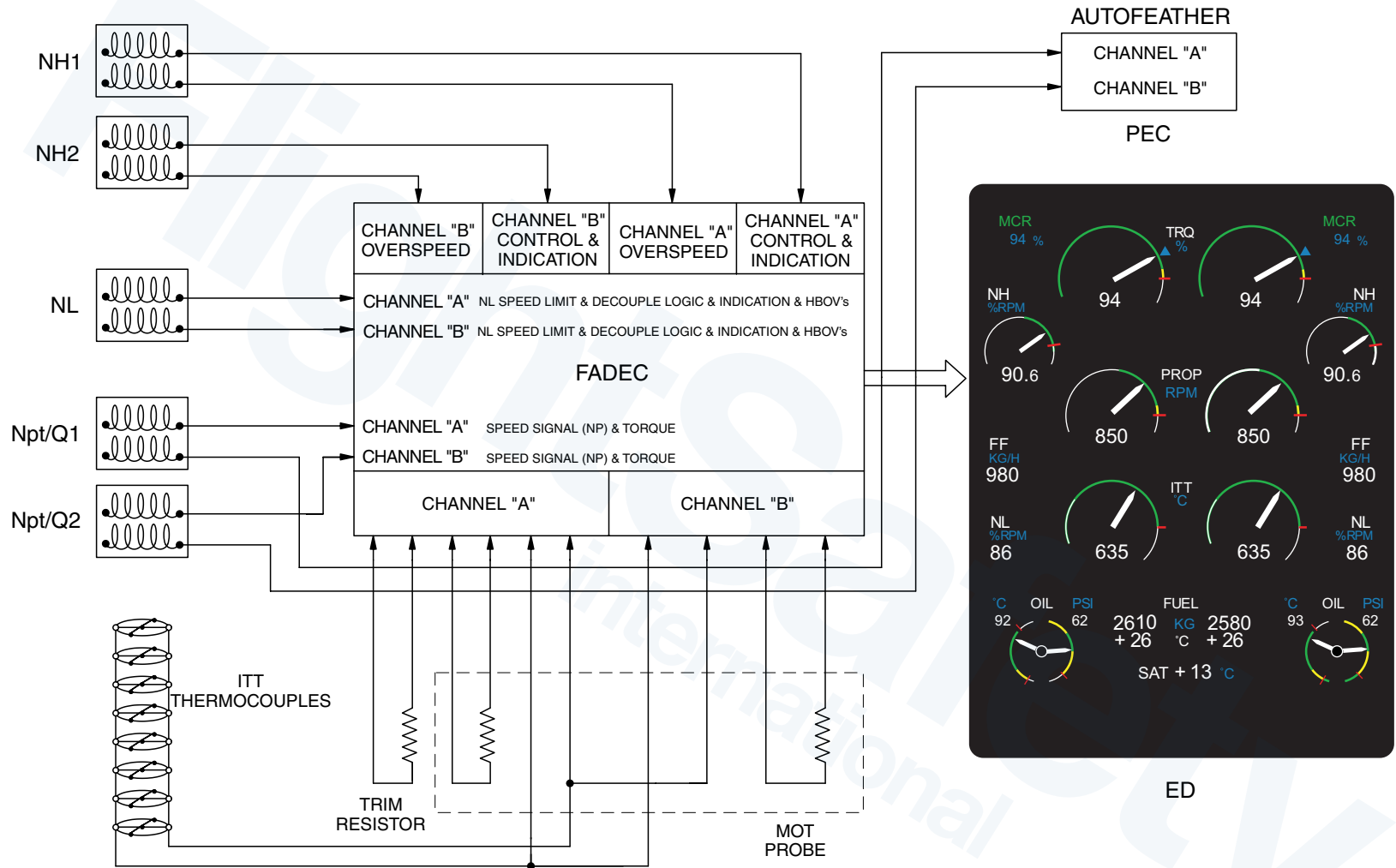
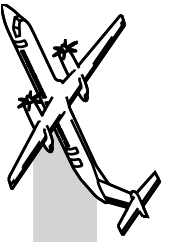
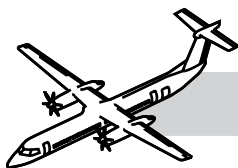


Figure 77-2. Engine Power Indicating Schematic





77-11-00 ENGINE POWER INDICATING

INTRODUCTION

The power indicating system gives data on the speed of the major rotating assemblies and the torque output.

GENERAL

Refer to Figure 77-2. Engine Power Indicating Schematic.

The power indicating system has:

- High Pressure Rotor Speed (N_h) sensors
- Torque sensors (Q)
- Low Pressure Rotor Speed (N_L) sensor
- Engine Controls Wiring Harness.

SYSTEM DESCRIPTION

The speeds of the major rotating assemblies are monitored by magnetic pulse pick-up probes installed at various locations on the engine.

These probes sense:

- High pressure rotor speed (N_h)
- Low pressure rotor speed (N_L)
- Propeller speed (N_P).

Power turbine rotor speed (N_{PT}) is sensed by the torque sensor and derived to propeller speed (N_P).

Electromagnetic pulses are generated when associated gear teeth or lugs pass through the magnetic field created by a permanent magnet at the probe or sensor tip.

The pulse frequency is transmitted to the FADEC, processed and then sent to the ED.

The N_{PT}/Q signals are also sent to Propeller Electronic Control (PEC) for autofeather and uptrim control.

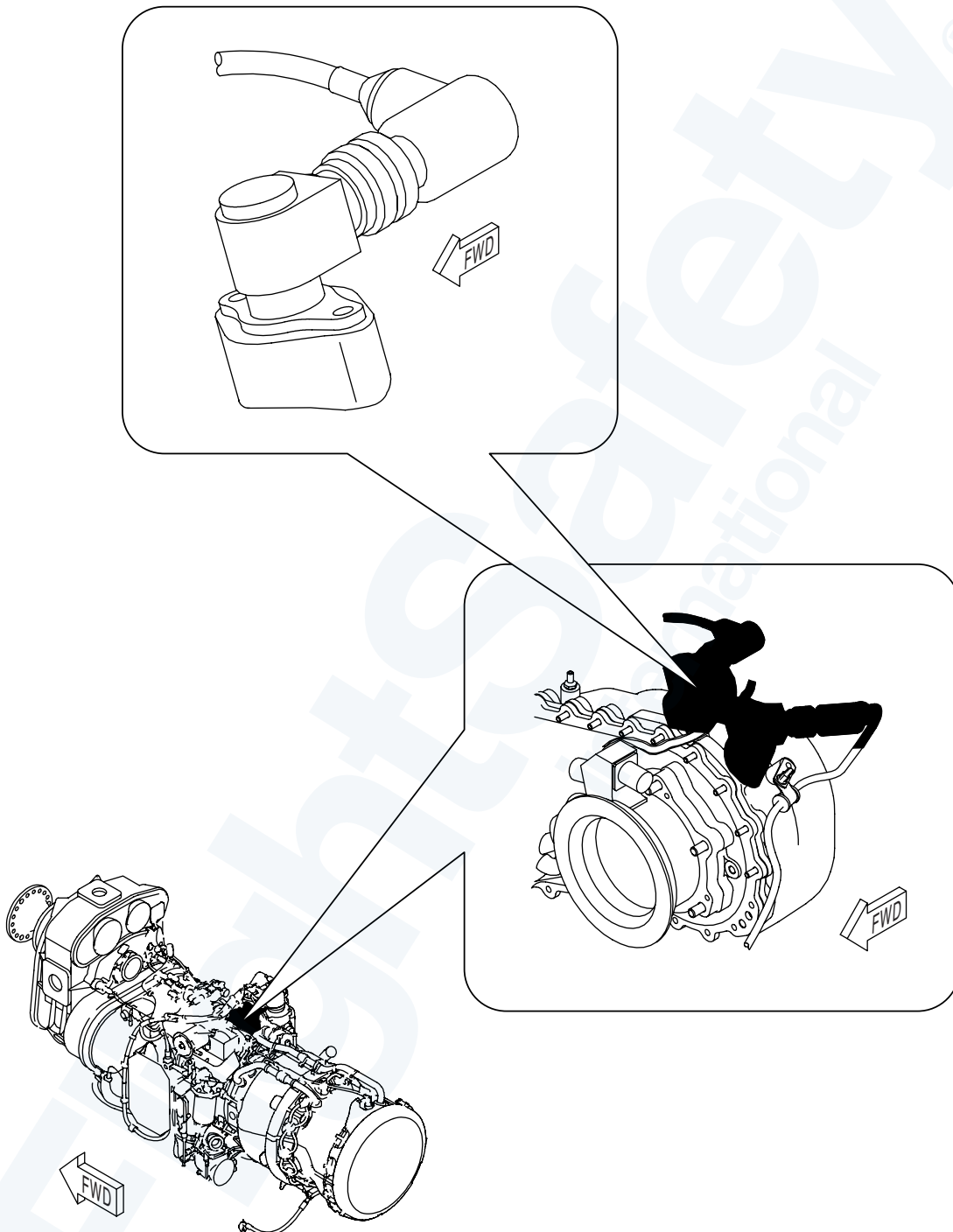
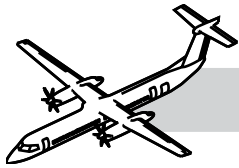
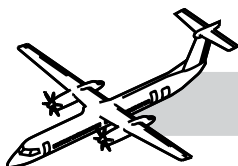


Figure 77-3. N_h Sensors



COMPONENT DESCRIPTION

NOTES

High Pressure Rotor Speed (N_h) Sensor

Refer to Figure 77-3. N_h Sensors.

The N_h sensor gives high pressure compressor speed information to the FADEC.

There are two N_h sensors on top of the Accessory Gearbox (AGB).

Each N_h sensor is a sealed unit and has two coils.

The sensors pick up high pressure rotor speed signals from the fuel pump/PMA gearshaft teeth.

One coil provides a N_h speed signal to the FADEC (for use in the control and indication logic).

The other coil provides a N_h speed signal for the overspeed logic.

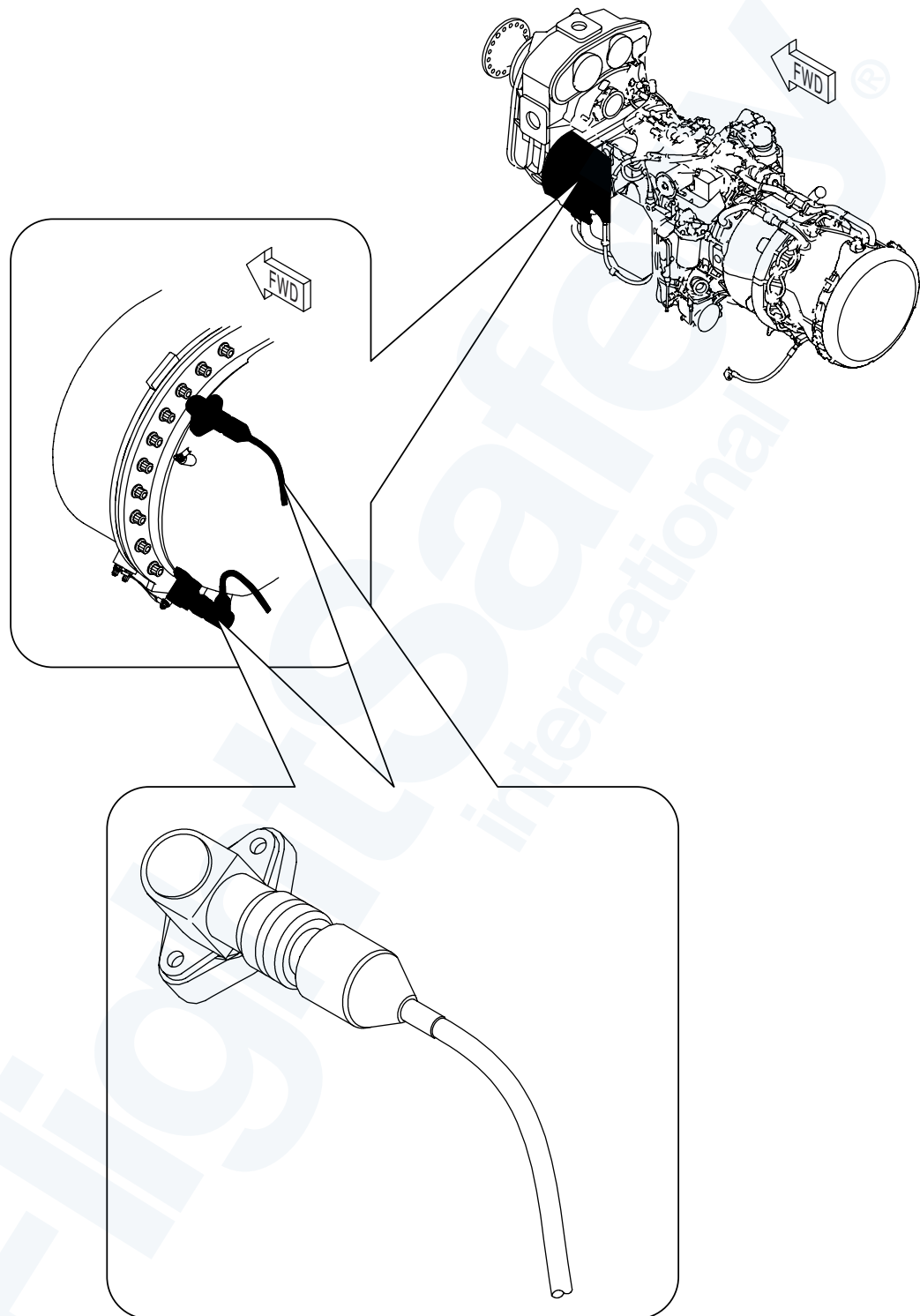
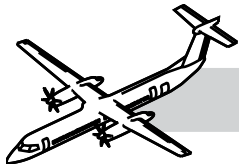
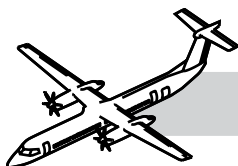


Figure 77-4. Torque Sensors



Power Turbine Speed N_{PT} Torque Sensor (Q)

NOTES

Refer to Figure 77-4. Torque Sensors.

The torque sensors send torque and power turbine speed information to the FADEC and PEC.

There are two torque sensors on the air inlet case.

The sensors take their signal from the left torque shaft in the reduction gearbox.

Correction of torque shaft stiffness caused by temperature change is calculated by the FADEC.

The Main Oil Temperature (MOT) probe (on top of the front inlet case) sends the temperature information for this computation.

Each sensor has two coils to supply information to FADEC and PEC:

- FADEC uses the Torque and N_{PT} speed signal for control logic
- PEC uses these signals for Autofeather logic, also its a reference signal to compare with signal from Magnetic Pickup Unit.

The Torque and N_{PT} speed signals are independent for channel A and B of both the FADEC and PEC.

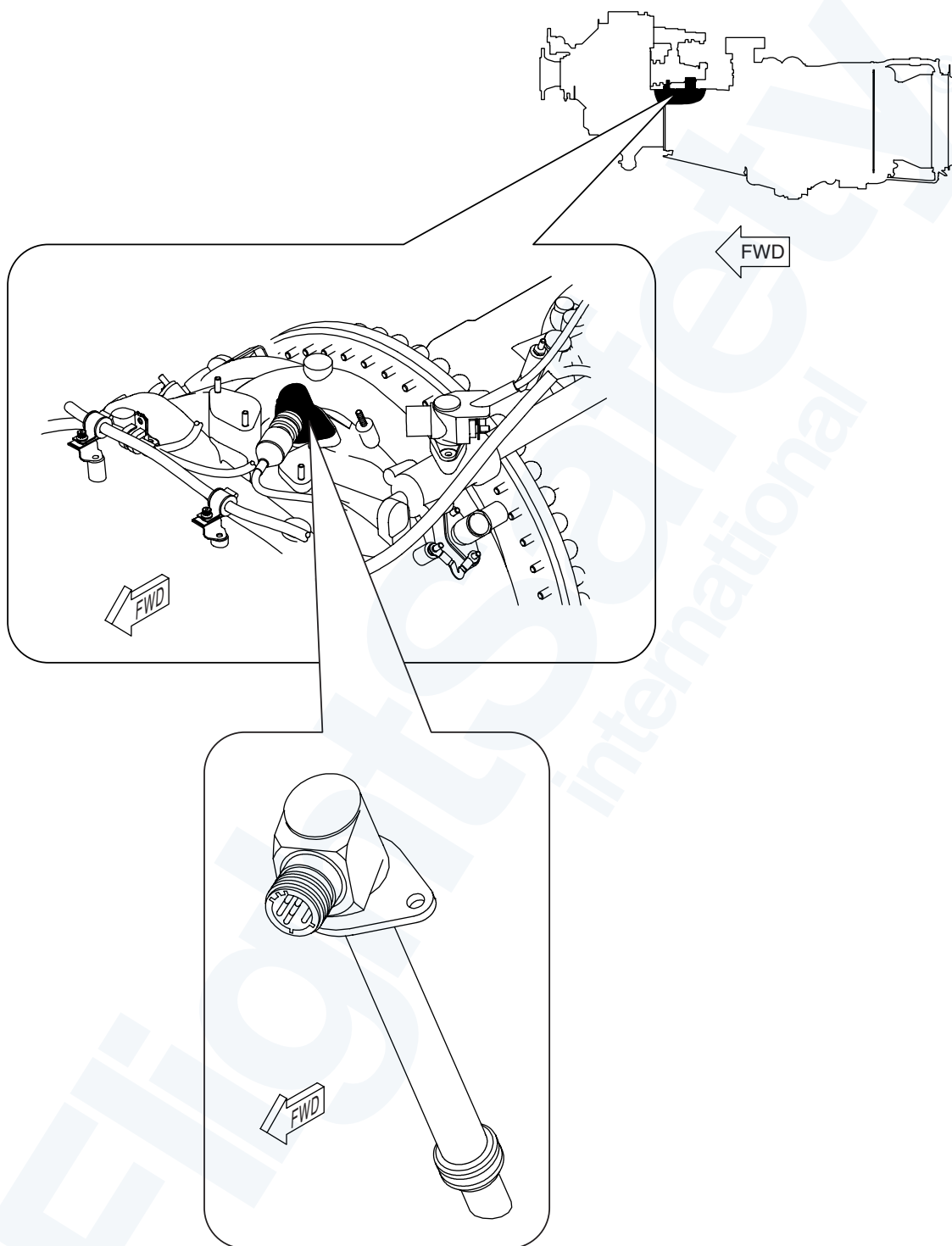
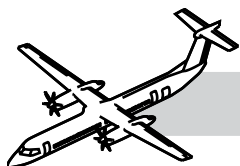
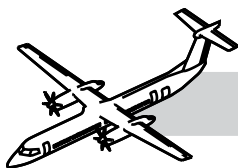


Figure 77-5. N_1 Sensor



Low Pressure Rotor Speed Sensor (N_L)

NOTES

Refer to Figure 77-5. N_L Sensor.

The N_L sensor sends low pressure compressor speed information to the FADEC.

There is one Low Pressure Rotor Speed Sensor on top of the front inlet case next to the MOT probe.

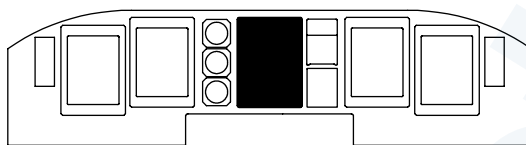
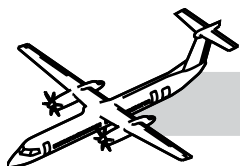
The N_L sensor is a sealed unit and has two coils.

The sensor picks up low pressure rotor speed from a toothed sleeve against the No.2.5 bearing.

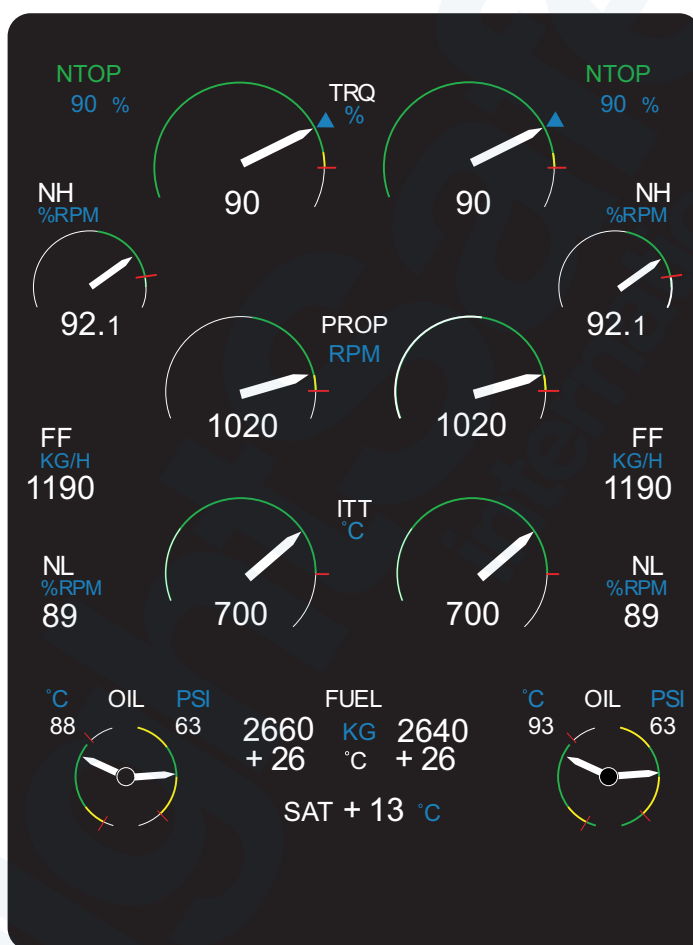
Each coil sends a N_L speed signal to channel A and B of the FADEC respectively.

N_L speed is used for:

- Controlling the Interstage and Intercompressor bleed valves
- Flight compartment indication
- The FADEC N_L speed limit logic and the N_L decouple logic.

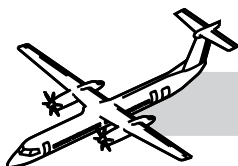


MAIN INSTRUMENT PANEL



77 ENGINE INDICATING

Figure 77-6. Engine Indication



CONTROLS AND INDICATION

NOTES

Refer to Figure 77-6. Engine Indication.

The following indications are displayed in both digital and analog format on the ED as percentages:

- High pressure compressor speed (N_h)
- Torque (Q)
- Propeller speed (N_p).

Low pressure compressor speed (N_L) is displayed in digital format only.

FADEC sends rating power information to the ED as indicated:

- By a torque bug
- On the torque analog display
- By digital display in the top left and right corners of the ED.

Selected Rating is indicated by an acronym above the rated torque digital display.

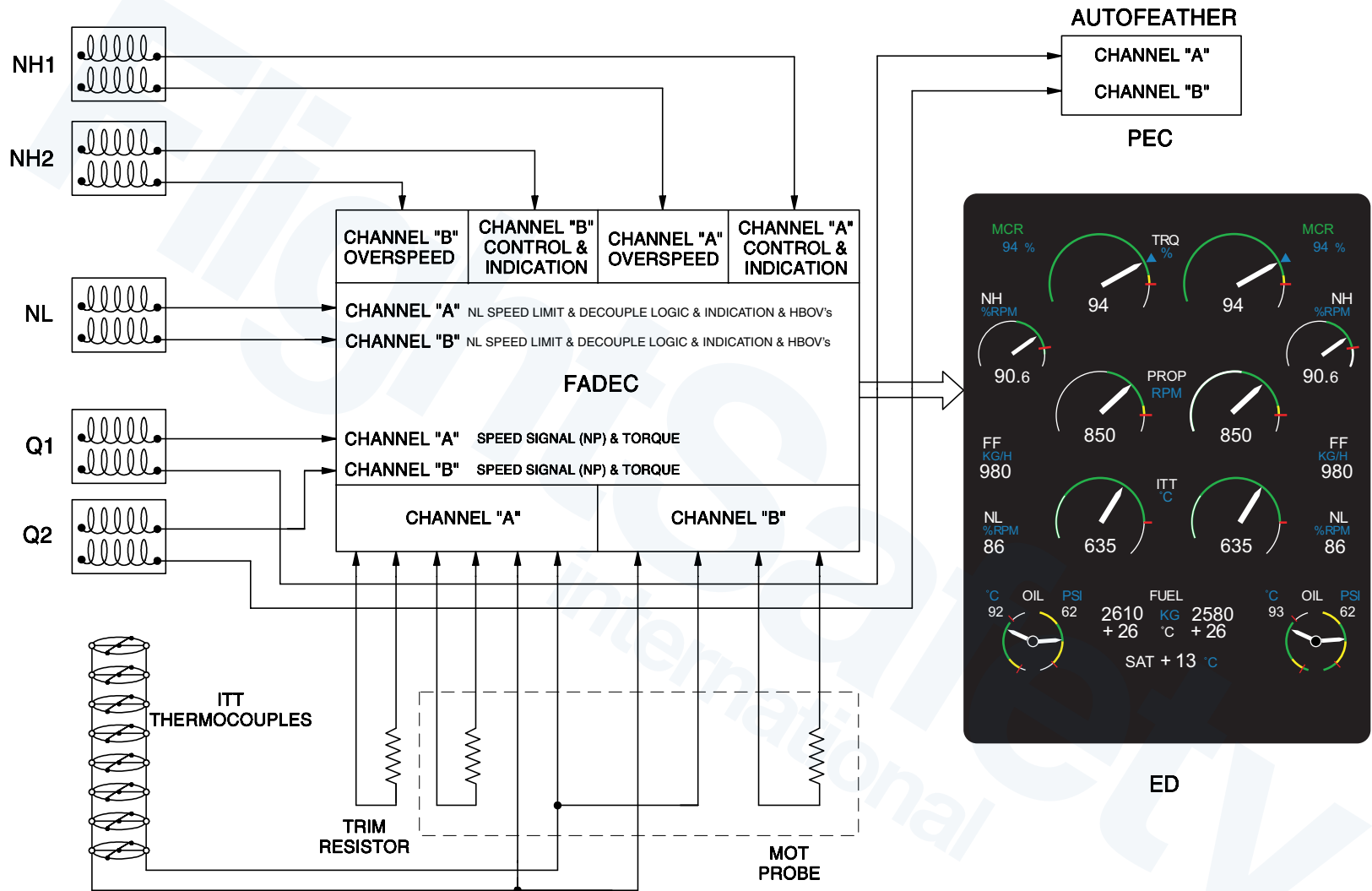
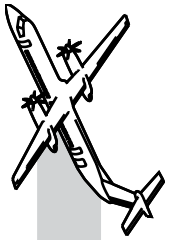
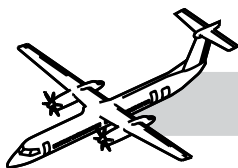


Figure 77-7. ITT Indicating Schematic





77-21-00 ENGINE TEMPERATURE INDICATING

NOTES

INTRODUCTION

The Engine Temperature Indicating System monitors engine performance.

GENERAL

The engine temperature indicating system supplies data on the Indicated-Turbine Temperature (ITT).

The system has:

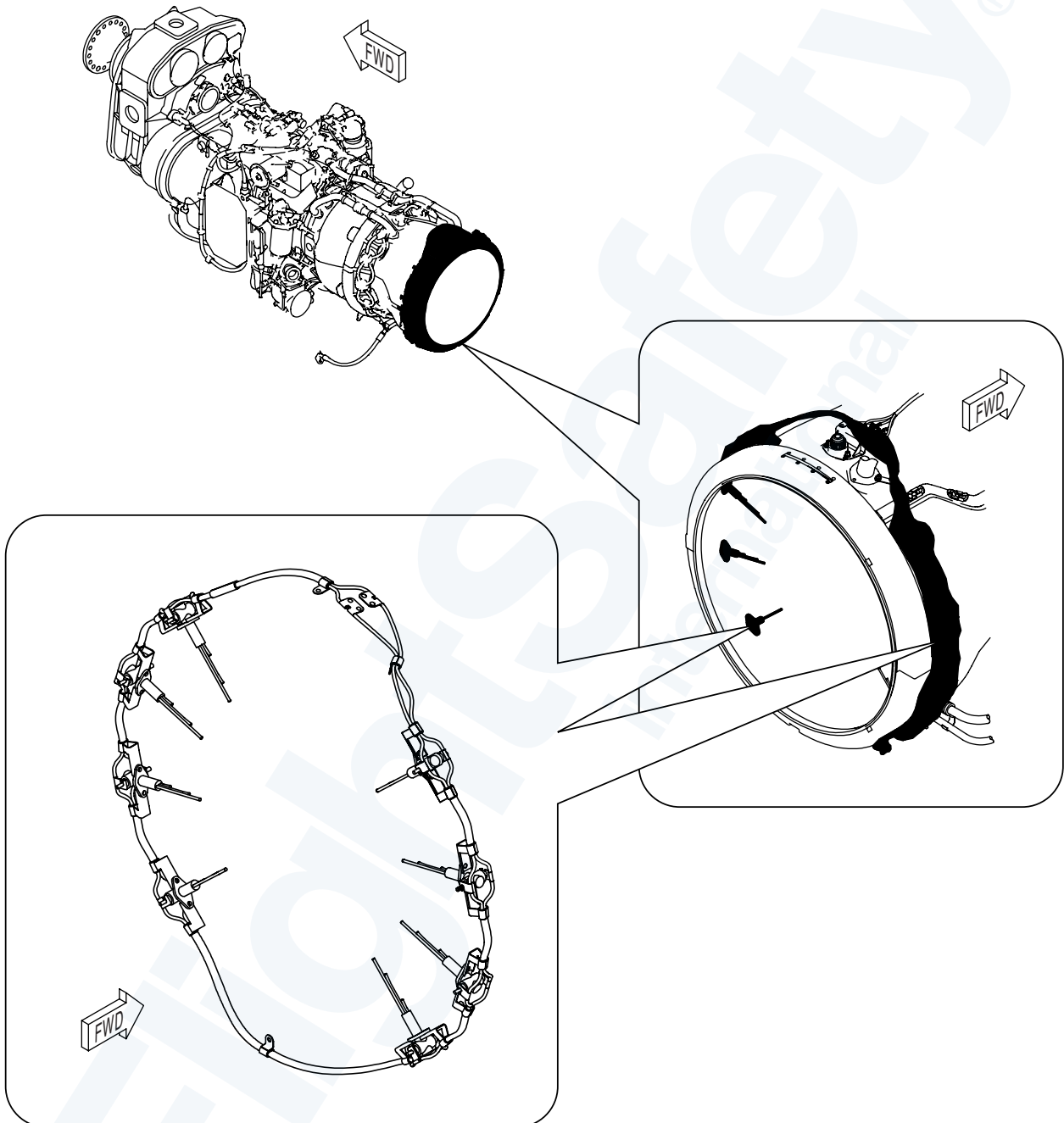
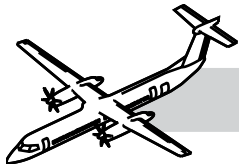
- Inter-turbine temperature probes.
- ITT Trim Resistor.

SYSTEM DESCRIPTION

Refer to Figure 77-7. ITT Indicating Schematic.

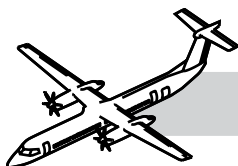
The T6 (ITT) probes send signals to the FADEC by the controls electrical wiring harness and the T6 wiring harness.

Each FADEC display channel (channel not in control) sends the signals on an ARINC 429 bus direct to the ED.



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Figure 77-8. ITT Probes



COMPONENT DESCRIPTION

Indicated Turbine Temperature (ITT) Probe

Refer to Figure 77-8. ITT Probes.

Eight probes consisting of three temperature sensing elements give 24 measurement points for comprehensive gas temperature measuring.

The sensing elements are located to give sufficient coverage for an accurate ITT. The probes are equi-spaced, on right and left sides, with no probes at top or bottom around the circumference at station 6 of the engine (downstream of the 2nd Power Turbine).

ITT measurement is done by 24 chromel-alumel thermocouples connected in parallel by the ITT wiring harness.

The total signal is connected at a junction at the top rear of the engine.

The junction is connected to a wiring harness which goes to the Main Oil Temperature/Cold Junction sensor.

The trimmed ITT and Main Oil Temperature (MOT) are processed by the FADEC and output to the flight compartment and EMU over an ARINC 429 bus.

ITT IMMERSION THERMOCOUPLE INDICATION

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Use a Barfield Digital Turbine Temperature Tester or equivalent to do tests.

For an engine that is at greater than ambient temperature as follows:

- Test resistance from the ITT junction to ground, must be >200 kilohms
- Test insulation resistance of each thermocouple, must be >200 kilohms
- Test for open circuit ITT immersion thermocouples.

The engine must be at ambient air temperature when you do this check.

- Test resistance of each thermocouple, connect the positive lead to the chromel terminal and the negative lead to the alumel terminal of the ITT thermocouple, must be 0.16 to 0.20 ohms. Calculate average of absolute values to get actual value of thermocouples.

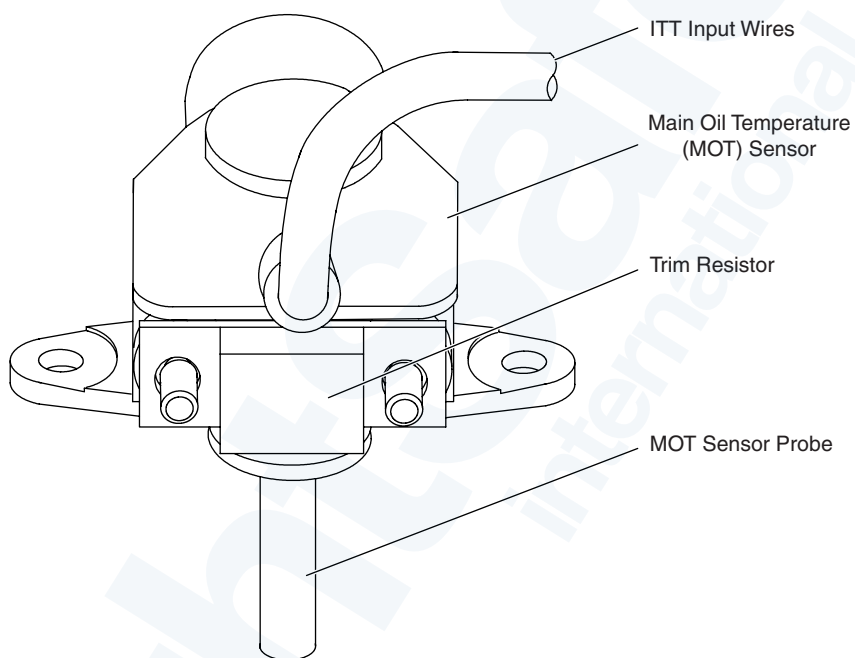
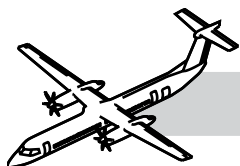
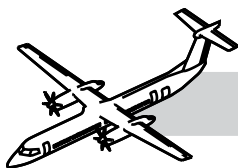


Figure 77-9. MOT Sensor and ITT Trim Resistor Detail



ITT Trim Resistor

NOTES

Refer to Figure 77-9. MOT Sensor and ITT Trim Resistor Detail.

The ITT Trim Resistor is on two stud terminals on the Main Oil Temperature/Cold Junction Sensor.

The average signal (by virtue of parallel arrangement) is routed through the ITT Trim Resistor and the cold junction reference before being received and processed by the FADEC.

The ITT Trim Resistor is used to provide compensation to the raw ITT signal.

The value of the resistor is set during engine pass off testing at the manufacturer or approved overhaul facility and is shown on the Engine Data Plate.

Training Information Points

Replace failed Lug Mounted Resistors before replacement of the EEC of the FADEC. The new FADEC will immediately ignore a failed resistor and use the trim value stored in its memory. This value will not be the correct value for the engine. Engine performance will be compromised.

OBEY ALL THE ELECTRICAL/ELECTRONIC, AND ELECTROSTATIC DISCHARGE SAFETY PRECAUTIONS WHEN HANDLING THE FADEC.

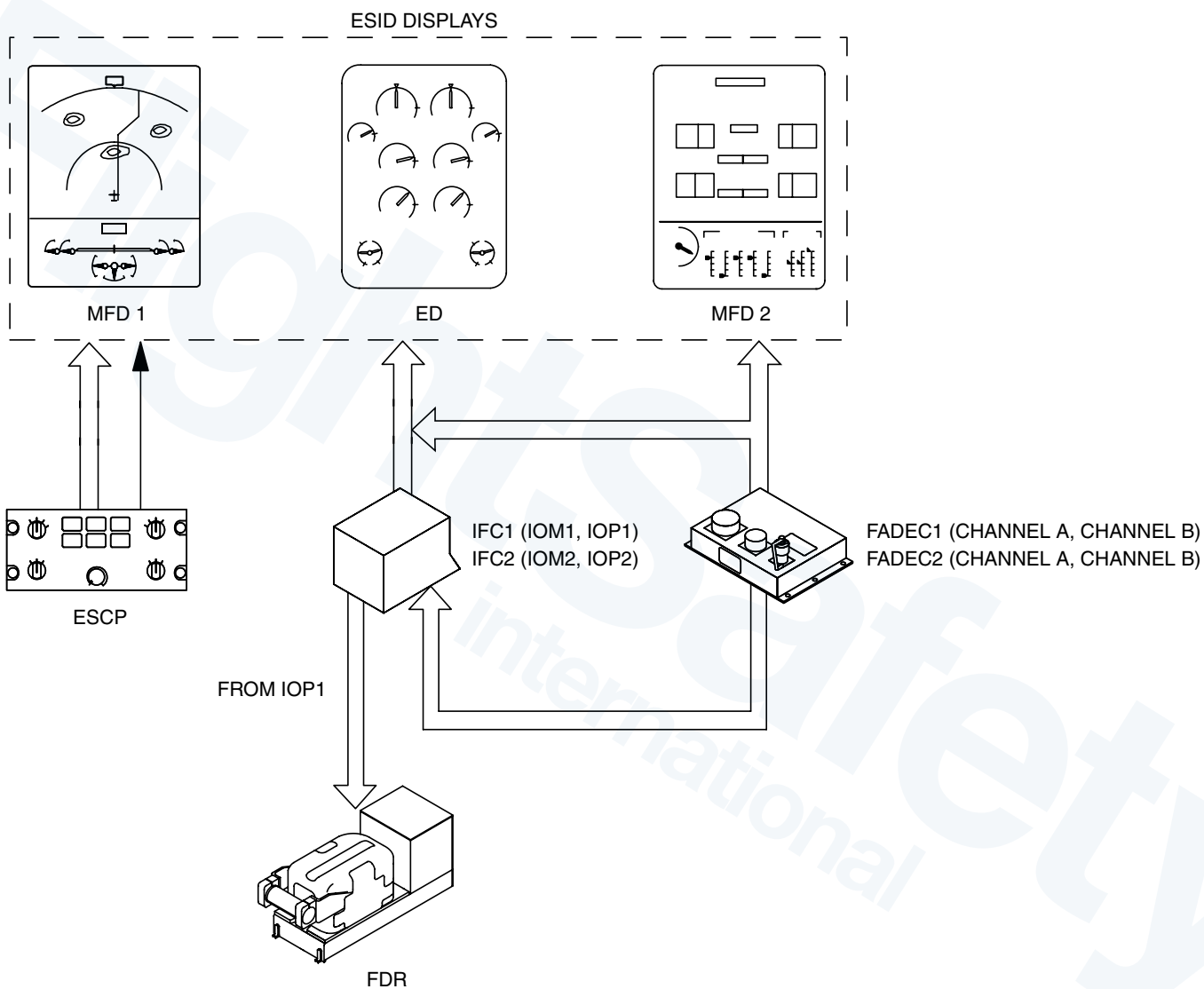
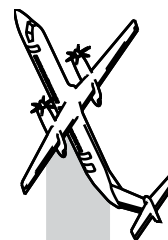
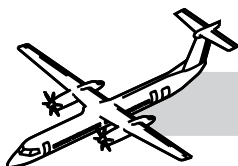


Figure 77-10. FADEC ARINC 429 Indicating System



77-30-00 ANALYZERS

INTRODUCTION

The system displays most of the engine parameters, plus discrete indications, in the flight compartment. It also sends powerplant information to the engine monitoring system and to the digital flight data recorder.

GENERAL

Refer to Figure 77-10. FADEC ARINC 429 Indicating System.

The function of the analyzers is done by the Engine Monitoring System (EMS) and the FADEC ARINC 429 System.

The EMS can record, display and download:

- FADEC and PEC fault and condition codes
- Engine Condition Trend Monitoring (ECTM) data
- Power assurance requirements and data
- Powerplant limit exceedance data
- Snapshot and transient powerplant and aircraft data
- Powerplant and aircraft flights, cycles and hours.

SYSTEM DESCRIPTION

Each FADEC display channel (channel not in control) sends signals on an ARINC 429 bus direct to the ED.

The FADEC information is also sent to the IFC for processing and transmission to the DFDR.

These parameters are sent to the flight compartment by discretes and analogue signals independent of the FADEC ARINC system:

- FADEC Fail, Caution and PEC Caution as discretes, sent directly to the CAWP
- Propeller Ground Range as a discrete to the Advisory Display Unit (ADU) and then to the lights on the glareshield panel
- Main Oil Pressure, Fuel Flow and Fuel Temperature from the engine mounted sensors as analog signals to the IFC, and from there to the ED on an ARINC 429 DATA bus.

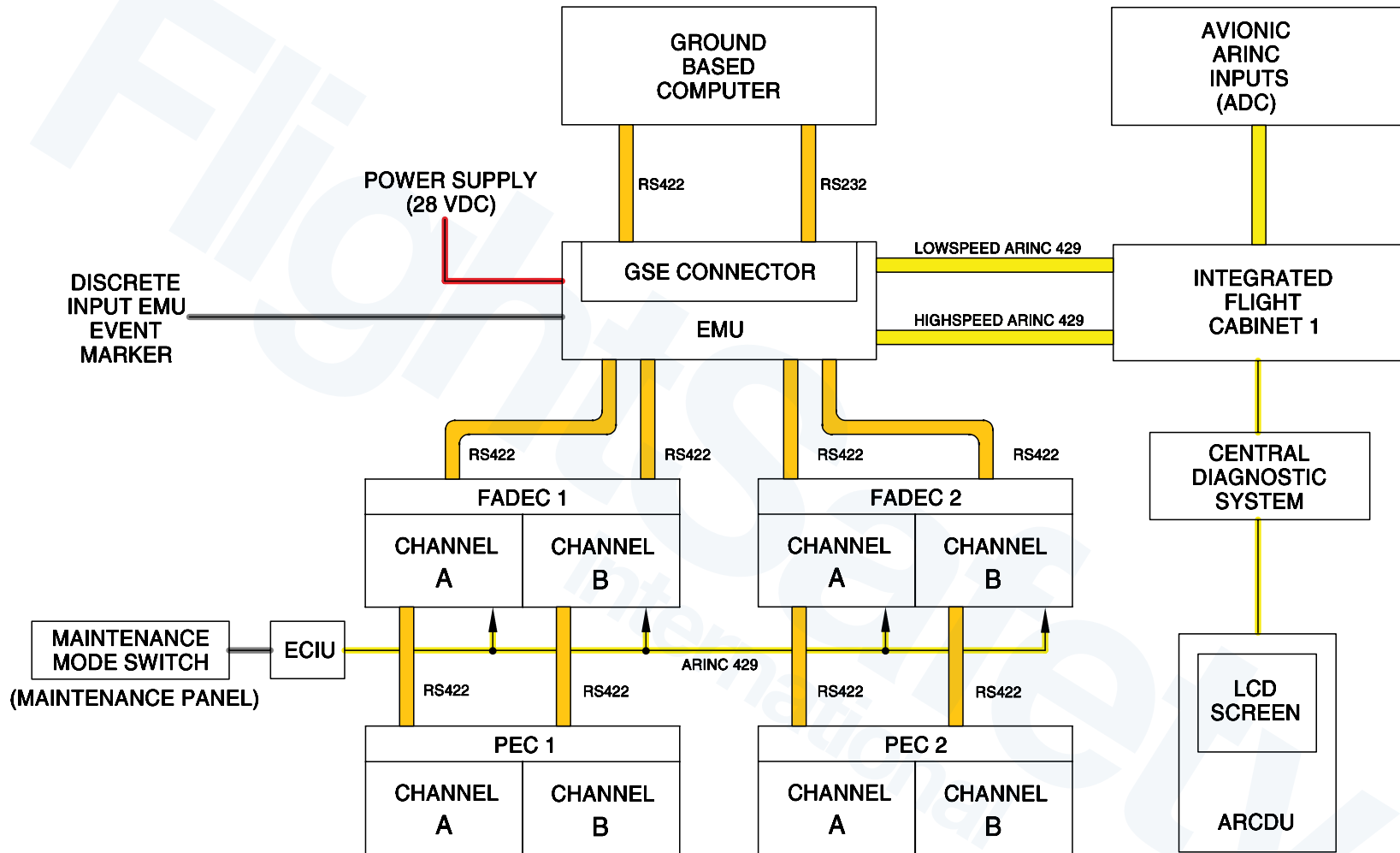
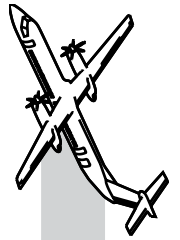
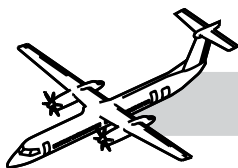


Figure 77-11. Engine Monitoring System (EMS) Schematic





77-31-00 ENGINE MONITORING SYSTEM

INTRODUCTION

The EMS collects data on every significant event that occurs during the operation of the engine.

GENERAL

Refer to Figure 77-11. Engine Monitoring System (EMS) Schematic.

The data collected by the EMS can be accessed in the aircraft through the ARCDU screen.

The data can also be downloaded to the laptop PC based Ground Based System (GBS).

The function of the Engine Monitoring System is performed by the:

- Engine Monitoring Unit (EMU)
- Ground Based System (GBS).

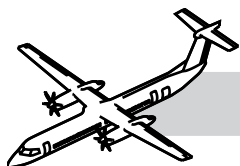
SYSTEM DESCRIPTION

The EMS has the following functions:

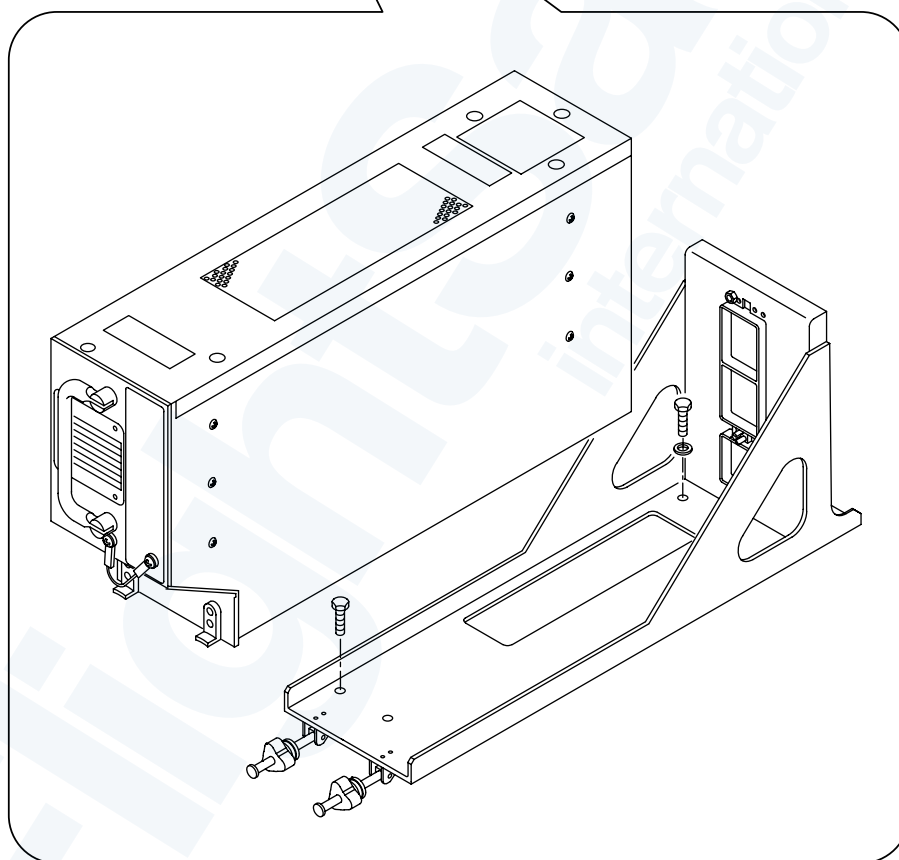
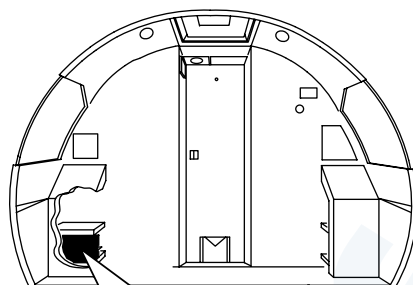
- Snapshot and/or Trace recording in response to an engine significant event (fault code, exceedance, etc.). A snapshot is a recording taken at the instant of the event and consists of 79 parameters generated by the FADEC/PEC. A Trace is a recording initiated at the instant of the event and consists of 49 parameters generated by the FADEC/PEC going back two minutes prior to the event and one minute after the event.
- Logging of Fault Codes generated by the FADEC and PEC.
- Recording of Engine Condition Trend Monitoring (ECTM).
- Averaging of ECTM conditions for the previous Flight Hours and alert to Maintenance if the trend deviates from the norm. Recording of time to

spooldown for high and low pressure spools. Alert to Maintenance if the spooldown time reduces below a minimum value.

- Monitoring of Engine parameters and logging any exceedance beyond operating limitations (Ref. AMM Chapter 5). These parameters include: ITT, Torque, $N_h/N_L/N_p$, Oil Pressure, Oil Temperature.
- Monitoring of Engine Health discretes, and logging any change in state. These discretes include: Chip Detectors, Low Oil Pressure, Oil Filter Impending Bypass, Low Fuel Pressure, and Fuel Filter Impending Bypass.
- Step by step Power Assurance procedure. Detailed results are available following successful completion.
- Live feedback of Flight Deck engine switch state. The position of any switch which provides an input to the FADEC or PEC will be displayed.
- Instructions to check the operation of Engine Health Discretes.
- Confirmation of powerplant trims: Torque Gain and Bias, ITT, configuration, BETA Feedback and Power Lever Angle (PLA) Feedback.
- Review of data stored in EMU memory. This feature allows the operator to view the date/time of each recording in memory, type of recording, and Flight Deck parameters at the time of the recording.
- Summary of aircraft and FADEC configuration. Aircraft Registration, Owner and Operator can be uploaded manually. Aircraft S/N is automatically uploaded from the CDS, FADEC, PEC, and EMU S/N are logged automatically. Hours and cycles accumulated on the above is available. EMU memory usage is available.

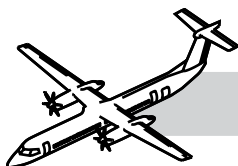


**FLIGHT COMPARTMENT BULKHEAD
VIEW LOOKING AFT**



77 ENGINE INDICATING

Figure 77-12. Engine Monitoring Unit (EMU)



Fault Retrieval

NOTES

Engine Monitoring Unit (EMU)

The EMU is in the flight compartment, on the side wall behind and below the First Officer's seat.

Refer to Figure 77-12. Engine Monitoring Unit (EMU).

CONTROLS AND INDICATIONS

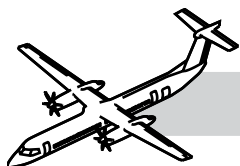
The data collected by the EMU can be viewed as follows:

- On aircraft on the ED N_h and torque gauges
- On aircraft on the ARCDU screen
- The data can be downloaded to a PC GBS.

Primary fault code display is on the torque gauge and N_h gauge. Channel A, FADEC and Lane A PEC fault codes are displayed on engine torque gauge. Channel B, FADEC and Lane B PEC fault codes are displayed on N_h Gauge.

From the main menu of the CDS, select EMU. The EMU menu will be displayed and any EMU function can be accessed. Note that where there is an active message waiting, the menu item color is amber rather than white.

This convention is carried through the sub-menus, the sub-sub-menus, and so on.



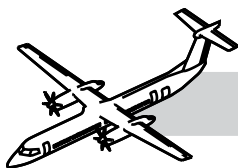
77-00-00 SPECIAL TOOLS & TEST EQUIPMENT

- PWC90004 Micro-ohmmeter
- Commercially Available Megohmmeter
- Simpson (or equivalent) Ohmmeter
- Commercially Available Joint, universal
- TT1200 or equivalent Barfield Digital Turbine Temperature Tester or equivalent
- PWC58104 Wrench, Mini-strap
- Glenair TG70 Wrench, Mini-strap
- Glenair TG69 Pliers, Soft-jawed
- DB9F to DB25M Cable, Download
- DB25F Connector/Backshell, EMU-clear Hardware Interlock
- DB9F to DB25M Modem Cable
- Commercially Available Computer, Laptop
- Commercially Available Diskette

77-00-00 MAINTENANCE PRACTICES

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

- FIM 77-11-00-810-803: Engine Indication, the torque indication is not correct - Fault Isolation.
- FIM 77-11-00-810-804: Engine 1 torque bug digital readout shows three white dashes (---), ED - Fault Isolation.
- FIM 77-11-00-810-808: Engine 1 torque bug digital readout shows three white dashes (---), MFD ENGINE system page - Fault Isolation.
- FIM 77-11-00-810-810: Engine 1 high pressure rotor speed indicator shows three white dashes (---), ED - Fault Isolation.
- FIM 77-11-00-810-812: Engine 1 high pressure rotor speed indicator shows three white dashes (---), MFD ENGINE system page - Fault Isolation.
- FIM 77-11-00-810-814: Engine 1 torque indicator shows three white dashes (---), ED - Fault Isolation.
- FIM 77-11-00-810-816: Engine 1 torque indicator shows three white dashes (---), MFD ENGINE system page - Fault Isolation
- FIM 77-11-00-810-818: Engine 1 low pressure rotor speed indicator shows three white dashes (---), ED - Fault Isolation.
- FIM 77-11-00-810-820: Engine 1 low pressure rotor speed indicator shows three white dashes (---), MFD ENGINE system page - Fault Isolation.
- FIM 77-11-00-810-822: Engine 1, ED, an indication discrepancy of the BLEED message - Fault Isolation.



- FIM 77-11-00-810-824: Engine 1, MFD, an indication discrepancy of the BLEED message - Fault Isolation.
- FIM 77-11-00-810-826: Engine 1, ED, an indication discrepancy of the MAINT message - Fault Isolation.
- FIM 77-11-00-810-828: Engine 1, MFD, an indication discrepancy of the MAINT message - Fault Isolation.
- FIM 77-11-00-810-831: Engine 1, ED, an indication discrepancy of the RDC TOP message - Fault Isolation.
- FIM 77-11-00-810-833: Engine 1, MFD, an indication discrepancy of the RDC TOP message - Fault Isolation.
- FIM 77-11-00-810-835: Engine 1, ED, an indication discrepancy of the NTOP message - Fault Isolation.
- FIM 77-11-00-810-837: Engine 1, MFD, an indication discrepancy of the NTOP message - Fault Isolation.
- FIM 77-11-00-810-839: Engine 1, ED, an indication discrepancy of the MTOP message - Fault Isolation.
- FIM 77-11-00-810-841: Engine 1, MFD, an indication discrepancy of the MTOP message - Fault Isolation.
- FIM 77-11-00-810-843: Engine 1, ED, an indication discrepancy of the MCL message - Fault Isolation.
- FIM 77-11-00-810-845: Engine 1, MFD, an indication discrepancy of the MCL message - Fault Isolation.
- FIM 77-11-00-810-847: Engine 1, ED, an indication discrepancy of the MCP message - Fault Isolation.
- FIM 77-11-00-810-849: Engine 1, MFD, an indication discrepancy of the MCP message - Fault Isolation.
- FIM 77-11-00-810-851: Engine 1, ED, an indication discrepancy of the MCR message - Fault Isolation.
- AMM 77-21-00-720-801: Functional Test of the Inter Turbine Temperature (ITT) System (continuity and resistance) (MRB#772100-201).
- AMM 77-21-01-720-801: Check for ITT Immersion Thermocouple Indication.
- AMM 77-21-02-720-801: Functional Test of the T1.8 Temperature Sensor.
- FIM 77-31-00-810-801: EMU General - Fault Isolation.
- AMM 77-31-01-470-801: Download of the Data from the EMU.
- AMM 77-31-00-710-802: Operational Test for Engine Fault Code Indications.